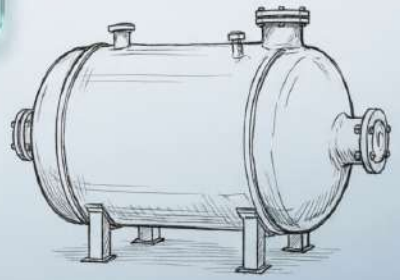
CORROTECH ITK LTD.

Hong Kong

GLASS-LINED / ENAMEL REACTORS

← Vertical • Horizontal →



- Jacketed
- Custom Design
- For Chemical, Pharmaceutical & Food Industries

www.pipefittings-hg.com

CorroTech ITK Ltd.

ABOUT CORROTECH ITK LTD.

CorroTech ITK Ltd. Hong Kong

CONTACT INFORMATION

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WhatsApp: +852 5344 5428

Website:
https://pipefittings-hg.com/

OUR EXPERTISE IN GLASS-LINED REACTORS:

- ✓ Direct partnerships with certified manufacturers (ISO 9001, ASME, PED)
- ✓ Custom engineering per DIN, ASME, GOST standards
- ✓ Materials: Glass 3008/3009, PTFE-lined, Hastelloy, Stainless Steel
- ✓ Full documentation: MTC 3.1, EN 10204, calculation reports, P&ID
- ✓ Third-party inspection available (SGS, BV, TUV, Lloyd's)
- ✓ Flexible payment via Hong Kong entities

CORE VALUE PROPOSITION

Hong Kong Registered Entity

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INDUSTRIES WE SERVE

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INDUSTRIES WE SERVE

- Chemical Processing
- Pharmaceutical & Fine Chemicals
- Food & Beverage Processing
- Agrochemicals & Fertilizers
- Specialty Chemical Synthesis
- High-corrosion Environment Installations

KEY APPLICATIONS:

Chemical Production & Processing
Acid production: Sulfuric, nitric, phosphoric, hydrochloric
Alkali processing: Caustic soda, ammonia, amines

Synthesis & Purification
Organic synthesis: Esters, ethers, aldehydes, ketones
Crystallization and purification processes

Hygienic & High-Purity Systems
CIP/SIP (Clean/Sterilize-In-Place) compatible systems
High-purity water treatment systems

Food & Agro Support Applications
Food additive production (citric acid, amino acids)
Flavor and fragrance manufacturing

Agrochemical Manufacturing
Pesticide and herbicidle synthesis
Fertilizer intermediates (Ammonium salts, phosphates)

Advanced Nutrient Solutions
Micronutrient chelate production

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TYPES OF GLASS-LINED REACTORS

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VERTICAL REACTORS

Standard configuration for most chemical processes

- Top-entering agitator with mechanical seal
- Full vacuum to +6 bar design pressure
- Volume range: 100L – 20,000L
- Jacket options: Dimpled, half-pipe, conventional
- Nozzle arrangements per customer P&ID

HORIZONTAL REACTORS

Ideal for viscous media and specific mixing requirements

- Side-entering or top-entering agitator options
- Enhanced heat transfer with extended surface area
- Volume range: 200L – 10,000L
- Suitable for crystallization, polymerization processes

JACKETED REACTORS

Precise temperature control via thermal media circulation

- Jacket types: Conventional, half-pipe coil, dimpled, external coil
- Heat transfer media: Steam, hot oil, thermal fluid, cooling water
- Design pressure: Full vacuum to +10 bar (jacket side)
- Optimized for exothermic/endothermic reactions

CUSTOM DESIGN SOLUTIONS

Tailored engineering for complex process demands

- Multi-jacket designs for complex temperature profiles
- Special nozzle configurations and instrumentation ports
- Reinforced structures for elevated pressure applications
- Integration with anciliary equipment (condensers, receivers)
- ATEX-compliant designs for explosive atmospheres

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VERTICAL REACTORS – DETAILED SPECIFICATIONS

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VOLUME & DIMENSIONS

Standard volumes: 100L, 200L, 500L, 1000L, 2000L, 3000L, 5000L, 10000L, 20000L

- Custom volumes available upon request
- **Aspect ratio (H/D):** 1.0 – 2.5 (optimized per process)
- **Nozzle sizes:** DN25 – DN300 (per P&ID requirements)

PRESSURE & TEMPERATURE

Vessel & Jacket

Design pressure (vessel):
Full vacuum to +6 bar (standard)

Design pressure (jacket):
Full vacuum to +10 bar (half-pipe)

Glass Linings

Design temperature:
-20°C to +200°C (Glass 3008)

Design temperature:
-30°C to +230°C (Glass 3009)

AGITATION SYSTEMS

Engineered for viscosity, mixing intensity and seal integrity

- **Types:** Anchor, paddle, turbine, propeller, retreat curve impeller
- **Power range:** 0.75 kW – 75 kW (per viscosity requirements)
- **Shaft sealing:** Mechanical seal (single/double), gland packing
- **Speed range:** 30 – 150 rpm (fixed or variable frequency drive)

JACKET CONFIGURATIONS

- **Conventional jacket:** Simple construction, moderate heat transfer
- **Half-pipe coil jacket:** High-pressure capability, efficient thermal media flow
- **Dimpled jacket:** Lightweight, cost-effective for low-pressure services
- **External coil:** Additional heating/cooling capacity, easy maintenance

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HORIZONTAL & CUSTOM REACTORS – SPECIFICATIONS

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HORIZONTAL REACTOR FEATURES

Ideal for high-viscosity media and specific mixing patterns

- Side-entering agitator reduces shaft deflection
- Enhanced bottom discharge for slurry and crystallizing services
- Volume range: 200L – 10,000L
- Design pressure: Full vacuum to +4 bar (standard)
- Design temperature: -20°C to +200°C

CUSTOM CONFIGURATION OPTIONS

- Multi-jacket designs for independent temperature zones
- Half-pipe coil jackets for high-pressure thermal media
- Dimpled jackets for efficient heat transfer at lower pressures
- External coils for additional heating/cooling capacity
- Reinforced nozzles for heavy instrumentation or piping loads
- Special linings: PTFE-backed glass, tantalum patches for extreme conditions

ENHANCED SAFETY FEATURES

- ✓ Rupture discs and pressure relief valves
- ✓ Temperature and pressure monitoring ports
- ✓ Emergency cooling connections
- ✓ Grounding provisions for static dissipation
- ✓ Explosion-proof motor and instrumentation options (ATEX/IECEx)

SPECIAL SERVICE CAPABILITIES

- High-pressure synthesis: Reinforced design up to +10 bar
- Gas-liquid reactions: Enhanced gas dispersion agitators
- Crystallization services: Scraped-surface agitator options
- Polymerization: High-torque agitation with cooling capacity

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GLASS ENAMEL TYPES & MATERIAL COMPATIBILITY

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GLASS 3008 (STANDARD)

Excellent resistance to most acids (except HF and hot phosphoric)

- Good chemical durability for general chemical processing
- Operating temperature range: -20°C to +200°C
- Thermal shock resistance: $\Delta T \leq 110°C$
- Applications: Acid production, organic synthesis, pharmaceutical intermediates

GLASS 3009 (THERMAL SHOCK RESISTANT)

Enhanced thermal shock resistance: $\Delta T \leq 150°C$

- Improved mechanical strength and impact resistance
- Suitable for processes with frequent temperature cycling
- Operating temperature range: -30°C to +230°C
- Applications: Batch processes with heating/cooling cycles, crystallization

PTFE-LINED COMPONENTS (SUPPLEMENTARY)

For extreme chemical resistance beyond glass capabilities

- Nozzles, agitator blades, dip pipes available with PTFE lining
- Temperature limit: -20°C to -180°C
- Applications: Hydrofluoric acid service, hot concentrated alkalis

MEDIA COMPATIBILITY GUIDE

- **Acids (HCl, HNO_3, H_2SO_4 dilute):** Glass 3008/3009 recommended
- **Caustic solutions (NaOH, KOH):** Glass 3009 preferred for concentrated/hot service
- **Organic solvents:** Glass 3008 suitable for most applications
- **Halogenated compounds:** Verify compatibility; PTFE lining may be required
- **Hydrofluoric acid:** Not compatible with glass; PTFE-lined or Hastelloy alternatives

CORROSION ALLOWANCE

- Glass lining provides complete barrier protection; no corrosion allowance required
- Steel substrate designed per pressure vessel codes with standard allowances
- Critical areas (nozzles, agitator shaft) receive additional inspection

TECHNICAL SPECIFICATIONS – COMPARISON TABLE

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COMPARISON TABLE

	VERTICAL	HORIZONTAL	CUSTOM
Volume Range	100L – 20,000L	200L – 10,000L	Custom per request
Design Pressure (vessel)	Vacuum to +6 bar	Vacuum to +4 bar	Up to +10 bar (reinforced)
Design Pressure (jacket)	Vacuum to +10 bar	Vacuum to +6 bar	Custom per thermal media
Design Temperature	-30°C to +230°C	-20°C to +200°C	Per glass type selection
Glass Type Options	3008, 3009	3008, 3009	3008, 3009, PTFE-lined
Agitator Types	Anchor, turbine, paddle	Side-entering turbine, anchor	Custom per mixing duty
Seal Options	Mechanical (single/double), gland	Mechanical seal preferred	Per API 682 guidelines
Nozzle Configuration	Top/side per P&ID	Side/bottom optimized	Fully customizable
Jacket Types	Conventional, half-pipe, dimpled	Conventional, half-pipe	Multi-zone, external coil
Special Features	Standard instrumentation ports	Enhanced discharge design	ATEX, high-pressure, multi-jacket

SELECTION CRITERIA

- **Process type:** Batch vs. continuous, reaction kinetics
- **Media characteristics:** Viscosity, abrasiveness, corrosivity
- **Thermal requirements:** Heating/cooling duty, temperature control precision
- **Mixing intensity:** Low-shear blending vs. high-shear dispersion
- **Safety considerations:** Exothermic reactions, hazardous materials
- **Regulatory requirements:** GMP, FDA, ATEX, PED compliance

STANDARDS & COMPLIANCE

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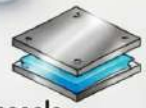
DESIGN & MANUFACTURING STANDARDS

- **DIN 28136:** Glass-lined steel equipment – Design, manufacture, testing
- **ASME Section VIII Div.1:** Pressure vessel code (on request)
- **PED 2014/68/EU:** Pressure Equipment Directive compliance
- **GOST R 53677 / GOST R 52857:** Russian standards for process equipment
- **EN 13445:** Unfired pressure vessels (European standard)
- **ATEX 2014/34/EU:** Equipment for explosive atmospheres (optional)

MATERIAL STANDARDS



- **ASTM A516 / EN 10028-2:** Carbon steel plates for pressure vessels
- **ASTM A240 / EN 10088-2:** Stainless steel plates (for nozzles, internals)
- **DIN 28136 Annex B:** Glass enamel quality requirements
- **ISO 15156 / NACE MR0175:** Materials for sour service (if applicable)

QUALITY & TESTING REQUIREMENTS

- ✓ 100% spark testing of glass lining at 25 kV (per DIN 28136)
- ✓ Hydrostatic pressure test at 1.3–1.5x design pressure
- ✓ Dimensional inspection per approved drawings
- ✓ Visual inspection of glass surface for defects, bubbles, inclusions
- ✓ Agitator run-in test (4 –8 hours) to verify mechanical performance

CERTIFICATIONS AVAILABLE

- ISO 9001:2015 — Quality Management System
- ISO 14001:2015 — Environmental Management
- ISO 45001 — Occupational Health & Safety
- PED Category I–IV (as applicable per pressure/volume)
- ASME Section VIII Div.1 'U' Stamp (on request)
- DIN 28136 compliance certificate
- GOST-R / EAC Certification (for CIS markets)
- ATEX / IECEX certification for explosion-proof components

ENGINEERING SUPPORT & SELECTION PROCESS

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1. DATA COLLECTION

- **Reaction type:** Batch, continuous, catalytic, polymerization
- **Media composition:** Acids, alkalis, solvents, slurries, halogenated compounds
- **Operating conditions:** Temperature profile, pressure range, residence time
- **Agitation requirements:** Viscosity (1–100,000 cP), mixing intensity, shear sensitivity
- **Safety & regulatory inputs:** Exothermic potential, hazardous materials, GMP/FADA/ATEX compliance scope

2. PRELIMINARY DESIGN & SELECTION

- **Reactor type recommendation:** Vertical / Horizontal / Pressure-rated
- **Volume calculation:** Base process volume + 20% safety margin
- **Material selection:** Glass 3008 / 3009 / PTFE-lined; steel grade per DIN/ASME
- **Agitator specification:** Type (anchor/turbine/retreat curve), power (0.75–75 kW)
- **Jacket estimation:** Type (dimpled/half-pipe/conventional), heat transfer area (m^2)

3. DETAILED ENGINEERING

- **Mechanical design:** Fully compliant with DIN 28136 and ASME Section VIII Div.1
- **Structural analysis:** Stress mapping, nozzle load verification, flange rating check
- **Agitator shaft design:** Critical speed calculation, deflection limits, bearing life assessment
- **Seal system:** API 682-compliant mechanical seal selection (Plan 52/53/72)
- **P&ID development:** Full instrumentation tagging, nozzle schedule, interface points with auxiliaries

4. DOCUMENTATION & APPROVAL

- **General arrangement drawing:** PDF & native DWG format, ISO-standard annotation
- **Technical data sheet:** All parameters – volume, pressure, temperature, materials, coatings
- **Material specification & ITP:** Heat number traceability, NDE methods, inspection hold points
- **Calculation report:** Pressure vessel (UG-23/UG-32), heat transfer (U-value, ΔT-LMTD), agitation torque
- **Commercial offer:** Validity period, lead time (12–26 weeks), payment terms (LC/T/T), delivery incoterm

DELIVERABLES & RESPONSE TIME

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WHAT WE DELIVER

- Technical proposal with process summary and key parameters
- General arrangement drawing with dimensions and nozzle schedule
- Material specification (DIN/ASME/GOST equivalents)
- Quality plan: inspection points, NDT requirements, documentation
- Commercial offer: pricing, payment terms, delivery schedule

RESPONSE TIME

Standard Timeline

Preliminary selection: within 48 hours

Detailed proposal: 5–7 business days (after data approval)

Urgent Requests

Expedited service available — lead time reduction up to 40% upon agreement

DOCUMENTATION INCLUDED WITH SHIPMENT

- Material Test Certificates (MTC) EN 10204 3.1 / 3.2
- Certificate of Conformity / Compliance
- Hydrostatic Test Report
- Spark Test Report (glass lining inspection)
- Dimensional Inspection Report
- Welding Procedure & Welder Qualification Records
- Agitator Run-in Test Report
- Packing List & Commercial Invoice
- Installation & Operation Manual (English/Russian/Chinese)
- As-built drawings and calculation summary (on request)

DELIVERY ASSURANCE

- ✅ All documents reviewed by QA engineer prior to release
- ✅ Electronic package delivered via secure client portal (PDF + native formats)
- ✅ Hard copies shipped in tamper-evident document case

TIMELINE TRANSPARENCY

- ✅ Real-time production tracking
- ✅ Weekly progress reports with photos, milestone notifications (engineering, production, QC, shipping)

QUALITY CONTROL PROCEDURES

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MANUFACTURER QUALIFICATIONS

- **ISO** 9001:2015 — Quality Management System
- **ISO** 14001:2015 — Environmental Management
- **ISO** 45001 — Occupational Health & Safety
- Welding Procedure Qualification (WPQR / WPS) per ASME Section IX
- Glass application certified per DIN 28136
- Authorized Inspection Agency (AIA) partnership with global accreditation

PRODUCT CERTIFICATIONS

- **PED 2014/68/EU — Category I to IV** (as applicable per pressure/volume)
- **ASME Section VIII** Div.1 **"U"** Stamp (on request)
- **DIN** 28136 compliance certificate
- **GOST-R / EAC Certification** (for CIS markets)
- **ATEX 2014/34/EU** certification for explosion-proof components
- **IECEx** certification for hazardous area equipment
- **NACE MR0175 / ISO 15156** compliance for sour service applications

THIRD-PARTY INSPECTION

- **SGS** — Available on request, full FAT witnessing & report issuance
- **Bureau Veritas (BV)** — Design review, material inspection, pressure testing
- **TUV Rheinland / TUV SUD** — PED conformity assessment & CE marking support
- **Lloyd's Register** — Independent verification of design, fabrication and QA processes
- **DNV GL** — Risk-based inspection planning & lifecycle integrity assurance
- **Client-appointed inspector** welcome — Dedicated coordination & access protocol

QUALITY CONTROL STEPS – DETAILED

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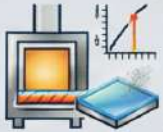
1. RAW MATERIAL INSPECTION

- Steel plate certification review (MTC 3.1)
- Chemical composition verification (spectrometer analysis)
- Mechanical properties testing (tensile, impact)
- Glass frit batch certification and traceability

2. FABRICATION & WELDING CONTROL

- WPS/PQR qualification per ASME IX / EN ISO 15614
- Welder qualification records maintained
- In-process dimensional checks at each fabrication stage
- Visual inspection (VT) of all welds pre-glassing

3. GLASS APPLICATION & FIRING



- Surface preparation: grit blasting to Sa 2.5 per ISO 8501-1
- Glass application: uniform coating thickness verification
- Firing cycle control: temperature profile monitoring per DIN 28136
- Post-firing inspection: visual check for defects, thickness measurement

4. FINAL INSPECTION & TESTING

- 100% spark testing at 25 kV per DIN 28136 Annex C
- Hydrostatic pressure test at 1.3–1.5x design pressure
- Agitator run-in test: vibration, noise, temperature monitoring
- Dimensional verification per approved GA drawing
- Documentation compilation and final review

INSPECTION TRACEABILITY

✅ All raw material heat numbers fully traceable from MTC to vessel ID

REFERENCE PROJECTS & CLIENT FEEDBACK

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INDUSTRIES & APPLICATIONS

- Chemical Processing Plants (Asia, Middle East)
- Pharmaceutical Manufacturing Facilities (Europe, India)
- Food Processing Industries (Southeast Asia)
- Agrochemical Production Units
- Specialty Chemical Synthesis
- High-corrosion Environment Installations

TYPICAL PROJECT SPECIFICATIONS

- Vertical Glass-Lined Reactor, 5,000L, Glass 3009, -1 +4 bar, 150°C, turbine agitator, half-pipe jacket
- Horizontal Enamel Reactor, 2,000L, Glass 3008, vacuum +2 bar, 180°C, anchor agitator, conventional jacket
- Custom Multi-Jacket Reactor, 10,000L, Glass 3009, dual-zone temperature control, ATEX Zone 1
- PTFE-Lined Nozzle Assembly for HF service, DN150, -1 +3 bar, 120°C

MATERIALS SUPPLIED

- Glass 3008 (Standard): **65%** of projects
- Glass 3009 (Thermal Shock): **25%** of projects
- PTFE-Lined Components: **7%** of projects
- Stainless Steel Internals: **3%** of projects

WHY CHOOSE CORROTECH FOR GLASS-LINED REACTORS?

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DIRECT MANUFACTURER ACCESS

- No middlemen — competitive pricing, faster lead times
- Direct coordination with certified glass-lining factories across Asia & Europe

ENGINEERING EXPERTISE

- DIN 28136 / ASME compliance, thermal calculation support
- Full mechanical design, stress analysis and P&ID integration

MATERIAL FLEXIBILITY

- Glass 3008, 3009, PTFE-lined, stainless steel internals — all available
- Material selection aligned with media corrosivity, temperature cycling & regulatory scope

FULL DOCUMENTATION

- MTC 3.1, PED, DIN, GOST — ready for customs & regulatory audits
- Bilingual (EN/CN) or trilingual (EN/RU/CN) documentation packages — no localization delay

QUALITY ASSURANCE

- 100% spark testing, hydrostatic test, third-party inspection available
- All QA records digitally archived with timestamped video logs and NCR tracking

FLEXIBLE PAYMENT

- Hong Kong entity — smooth international transactions
- Payment terms: T/T, LC, or HKD/USD settlement via licensed financial institutions

GLOBAL LOGISTICS

- CIF/FOB/DDP — we handle shipping to your facility
- Customs clearance support, export licenses, hazardous goods classification (if applicable)

FAST RESPONSE

- Technical proposal within 48 hours
- Commercial offer within 24 hours after specs approval

CONTACT US

CORROTECH ITK LTD. Hong Kong

CONTACT INFORMATION

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WhatsApp: +852 5344 5428

Website: https://pipefittings-hg.com/

Business Hours:
Monday – Friday: 9:00 – 18:00 (HKT)
Weekend: By appointment

REQUEST FOR QUOTATION:

- Reactor type (Vertical / Horizontal / Pressure)
- Required volume (L) and working volume
- Process media and concentration
- Operating pressure and temperature range
- Agitation requirements (viscosity, mixing intensity)
- Heating/cooling media and duty
- Material preference (glass 3008/3009, PTFE-lined, etc.)
- Applicable standards (DIN / ASME / GOST / PED)
- Delivery destination and required date

WE GUARANTEE:

✓ Monday – Friday: 9:00 – 18:00 (HKT)
✓ Weekend: By appointment